

Telco Customer Churn Analysis – Data Analysis Case Study Report

Summary

Customer churn is one of the most critical business challenges in the telecommunications industry. Retaining existing customers is often more cost-effective than acquiring new ones. This case study focuses on analyzing customer churn behavior using the Telco Customer Churn dataset to identify major factors influencing customer attrition and to provide actionable business recommendations.

The project was performed using Python libraries such as Pandas, NumPy, and Matplotlib within Google Colab. The analysis included data cleaning, exploratory data analysis (EDA), and visualization of customer behavior patterns.

The findings reveal that customers with month-to-month contracts, shorter tenure periods, electronic check payment methods, and fiber optic internet services showed significantly higher churn rates. On the other hand, customers with long-term contracts and longer service tenure were more likely to remain loyal.

The analysis provides meaningful insights that telecom businesses can use to improve customer retention strategies, enhance customer satisfaction, and reduce revenue loss caused by churn.

Background

In the telecom sector, customer churn refers to customers discontinuing their services with a company. Due to intense market competition and the availability of multiple service providers, telecom companies continuously face challenges in retaining customers.

Understanding the factors responsible for churn enables businesses to:

- Improve customer satisfaction
- Reduce customer acquisition costs
- Increase profitability
- Develop targeted retention strategies
- Predict future churn behavior

This case study aims to explore customer churn patterns through data analysis and derive valuable business insights.

Problem Statement

The telecom company experiences customer loss due to various service-related and demographic factors. The objective of this project is to:

- Analyze customer data
- Identify churn patterns
- Understand factors affecting customer retention
- Generate actionable insights for business decision-making

Objectives of the Study

The primary objectives of this project are:

1. To analyze customer churn distribution.
2. To identify the relationship between customer demographics and churn.
3. To study the impact of contract type, payment method, internet services, and tenure on churn.
4. To perform exploratory data analysis using Python.
5. To generate visual insights for business recommendations.

Tools and Technologies Used

Tool/Technology	Purpose
Python	Data analysis and visualization
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib	Data visualization
Google Colab	Development environment
CSV Dataset	Customer churn data source

Dataset Overview

The dataset used for this project contains customer-related information from a telecom company. It includes demographic details, subscription information, billing details, and churn status.

Important Features

Feature	Description
CustomerID	Unique customer identifier
Gender	Male/Female
Dependents	Whether customer has dependents
Tenure(Months)	Duration of customer relationship
InternetService	Type of internet service
Contract	Contract type
PaymentMethod	Customer payment method
MonthlyCharges	Monthly bill amount
TotalCharges	Total amount charged
Churn	Whether customer left the service

Methodology

The project followed a structured data analysis process:

Data Collection

The Telco Customer Churn CSV dataset was loaded into Google Colab using Pandas.

```
import pandas as pd
```

```
df = pd.read_csv('Telco-Customer-Churn.csv')
```

Data Understanding

The dataset structure, data types, and missing values were examined using:

```
print(df.info())  
print(df.isnull().sum())
```

This helped identify:

- Number of rows and columns
- Null values
- Data consistency
- Data types

Data Cleaning

Data cleaning steps included:

- Removing duplicate rows
- Checking missing values
- Verifying data consistency

```
df = df.drop_duplicates()
```

Exploratory Data Analysis and Findings

❖ Overall Churn Distribution

Key Findings

- A significant portion of customers remained with the company.
- However, a noticeable percentage of customers had churned.
- Churn represents a major business risk because it directly affects recurring revenue.

Business Insight

The company should focus on identifying high-risk customer groups and proactively implementing retention strategies.

❖ Contract Type vs Churn

Observations

- Customers with month-to-month contracts showed the highest churn rate.
- Customers with one-year and two-year contracts had much lower churn.

Interpretation

Long-term contracts create stronger customer commitment and reduce the likelihood of switching providers.

Business Recommendation

The telecom company should:

- Promote long-term contracts through discounts and loyalty benefits.
- Offer incentives for upgrading from month-to-month plans.

❖ Tenure vs Churn

Tenure analysis studied the relationship between customer duration and churn.

Observations

- Customers with shorter tenure were more likely to leave.
- Long-term customers showed higher retention rates.

Interpretation

New customers are at greater risk of churn due to:

- Initial dissatisfaction
- Service expectations
- Competitor offers
- Lack of loyalty

Business Recommendation

The company should introduce:

- Better onboarding experiences
- Early engagement programs
- Personalized offers for new customers

❖ Internet Service vs Churn

This analysis examined churn behavior based on internet service type.

Observations

- Fiber optic customers had higher churn rates compared to DSL users.
- Customers without internet services had lower churn.

Interpretation

Possible reasons include:

- Higher pricing for fiber services
- Service quality issues
- Customer expectations not being met

Business Recommendation

The company should:

- Improve fiber service quality
- Optimize pricing strategies
- Conduct customer satisfaction surveys
- Payment Method vs Churn

Customer churn was analyzed across different payment methods.

Observations

- Customers using electronic checks had higher churn.
- Automatic payment users showed relatively lower churn.

Interpretation

Customers using automatic payments may have:

- Greater trust in the company
- Higher service continuity
- Lower payment friction

Business Recommendation

The company should encourage customers to adopt:

- Auto-pay systems
- Digital payment subscriptions
- Reward-based payment programs

➤ Dependents vs Churn

The relationship between dependents and churn was analyzed.

Observations

- Customers without dependents showed higher churn.
- Customers with dependents were more likely to stay.

Interpretation

Customers with family responsibilities may prioritize service stability and reliability.

Business Recommendation

The company can introduce:

- Family plans
- Multi-user discounts
- Household bundles

Key Business Insights

The analysis generated several important business insights:

Insight	Business Impact
Month-to-month customers churn more	Need for long-term retention plans
New customers are high-risk	Improve onboarding and engagement
Fiber optic customers show high churn	Service improvement required
Electronic check users churn more	Promote automated payments
Customers with dependents stay longer	Develop family-oriented plans

Recommendations

Customer Retention Programs

- Offer loyalty rewards for long-term customers.
- Provide renewal discounts for yearly contracts.
- Launch customer engagement campaigns.

Improve Customer Experience

- Improve service quality for fiber customers.
- Reduce complaint resolution time.
- Enhance technical support.

Personalized Marketing

- Target customers with short tenure.
- Provide personalized offers.
- Recommend bundled services.

Challenges Faced

During the project, several challenges were encountered:

- Handling missing or inconsistent data
- Understanding customer behavior patterns
- Selecting meaningful visualizations
- Interpreting churn-related business factors

Future Scope

This project can be enhanced further using advanced analytics and machine learning.

Possible future improvements include:

- Churn prediction models using Logistic Regression or Random Forest
- Interactive dashboards using Power BI or Tableau
- Customer segmentation analysis
- Sentiment analysis from customer feedback
- Real-time churn monitoring systems

Conclusion

This case study successfully analyzed telecom customer churn using Python-based data analysis techniques. The project identified major factors affecting customer retention, including contract type, tenure, internet service, payment method, and dependents.

The findings indicate that customers with shorter tenure, month-to-month contracts, fiber optic services, and electronic check payments are more likely to churn. These insights provide valuable direction for telecom companies to improve customer retention and reduce revenue loss.

By implementing customer-focused strategies and predictive analytics in the future, telecom companies can significantly improve customer loyalty and long-term business performance.